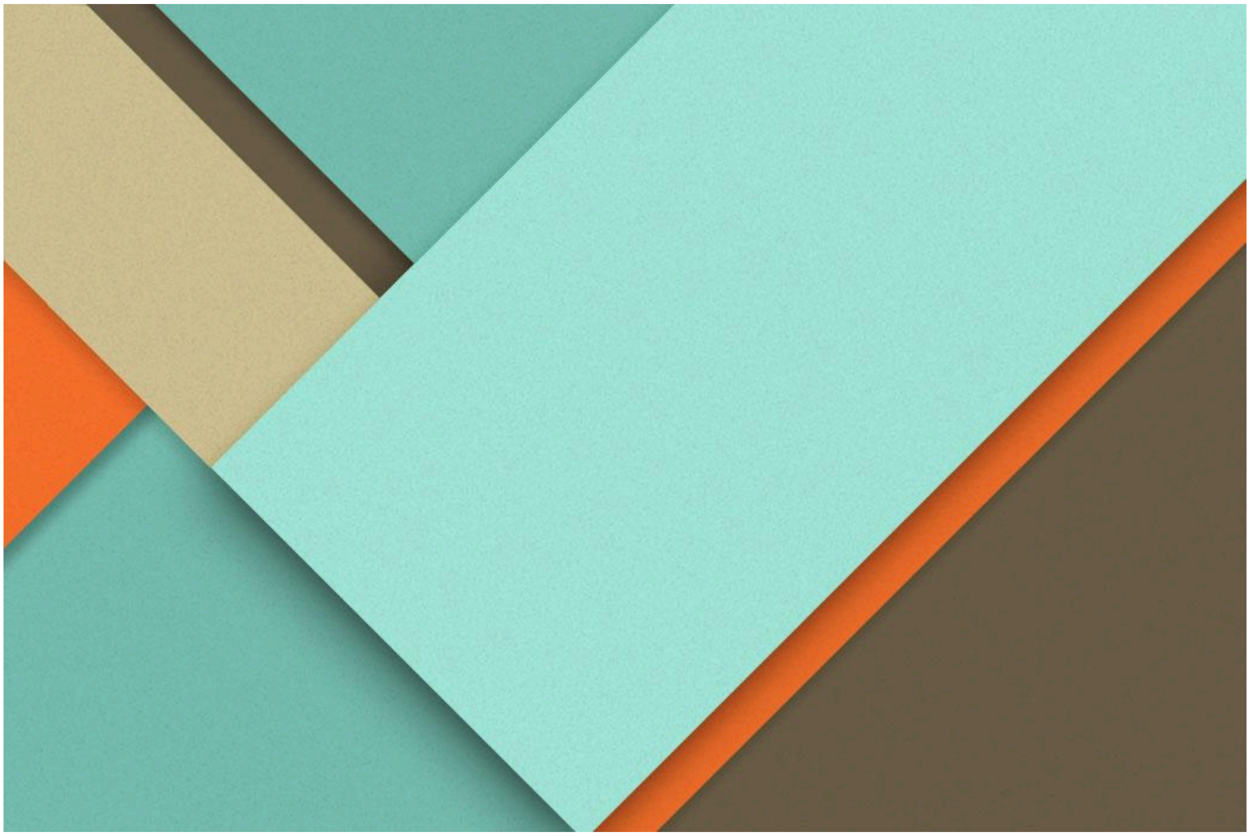

Mechatronic Linear Servo System

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Introduction

Mechatronic linear servo systems are integral components in modern automation and robotics, combining principles of mechanics, electronics, and computing to achieve precise motion control. These systems are designed to convert electrical signals into linear motion, providing accuracy, speed, and repeatability in applications ranging from industrial machinery to medical devices.

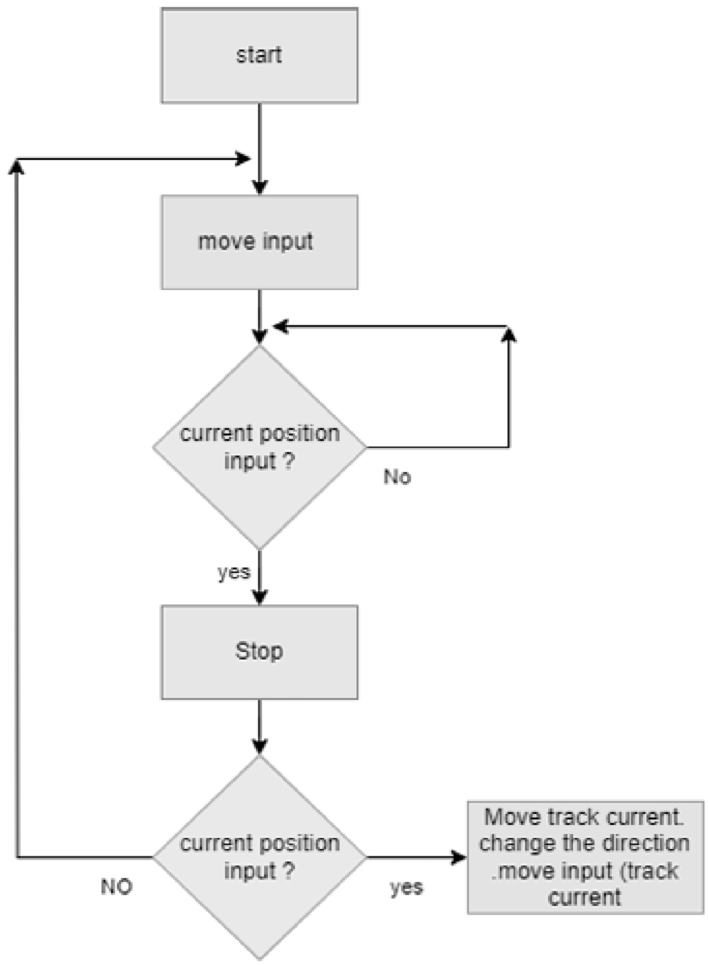
At the heart of a mechatronic linear servo system lies the servo motor, which drives a linear actuator to produce controlled movement along a straight path. This is achieved through a combination of feedback loops, sensors, and controllers, allowing the system to adapt to dynamic conditions and maintain performance. The integration of advanced software algorithms further enhances their versatility, enabling applications such as pick-and-place operations, CNC machining, and adaptive manufacturing.

The development of mechatronic linear servo systems is a testament to interdisciplinary engineering, as it relies on innovations in material science, control theory, and embedded systems. Their growing adoption is driven by the demand for automation solutions that offer both precision and reliability, ensuring efficiency in complex and repetitive tasks across various industries.

By leveraging the synergy of mechanical design and electronic intelligence, mechatronic linear servo systems exemplify the capabilities of modern engineering in shaping the future of automation and control systems.

Methodology

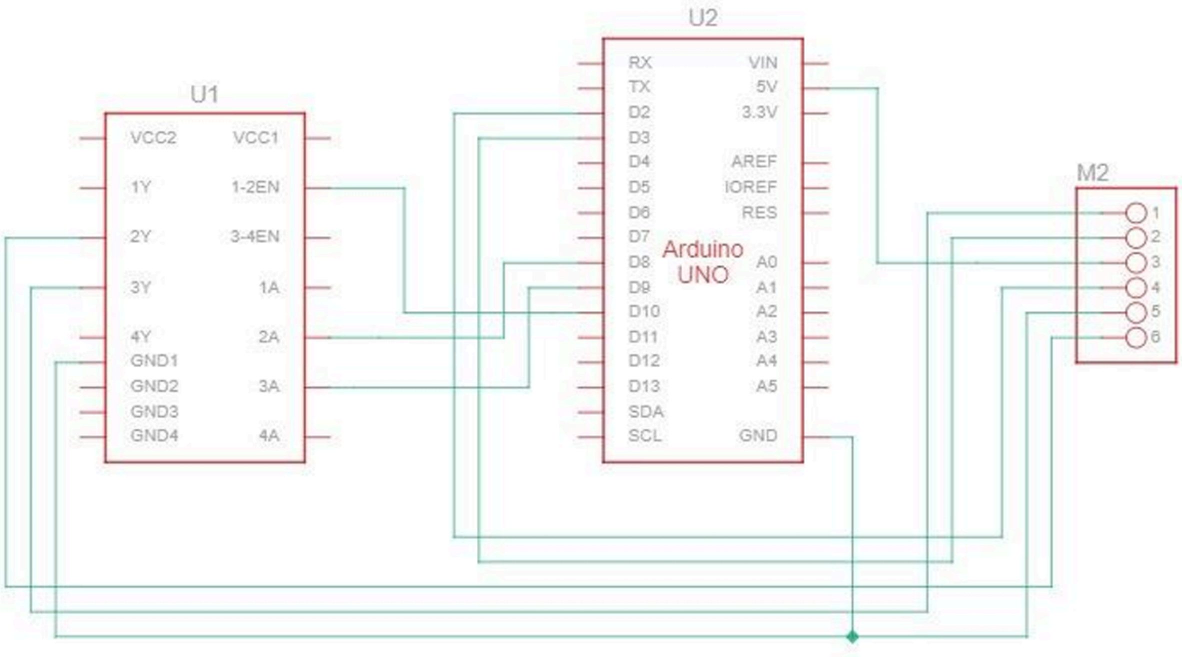
Workflow:

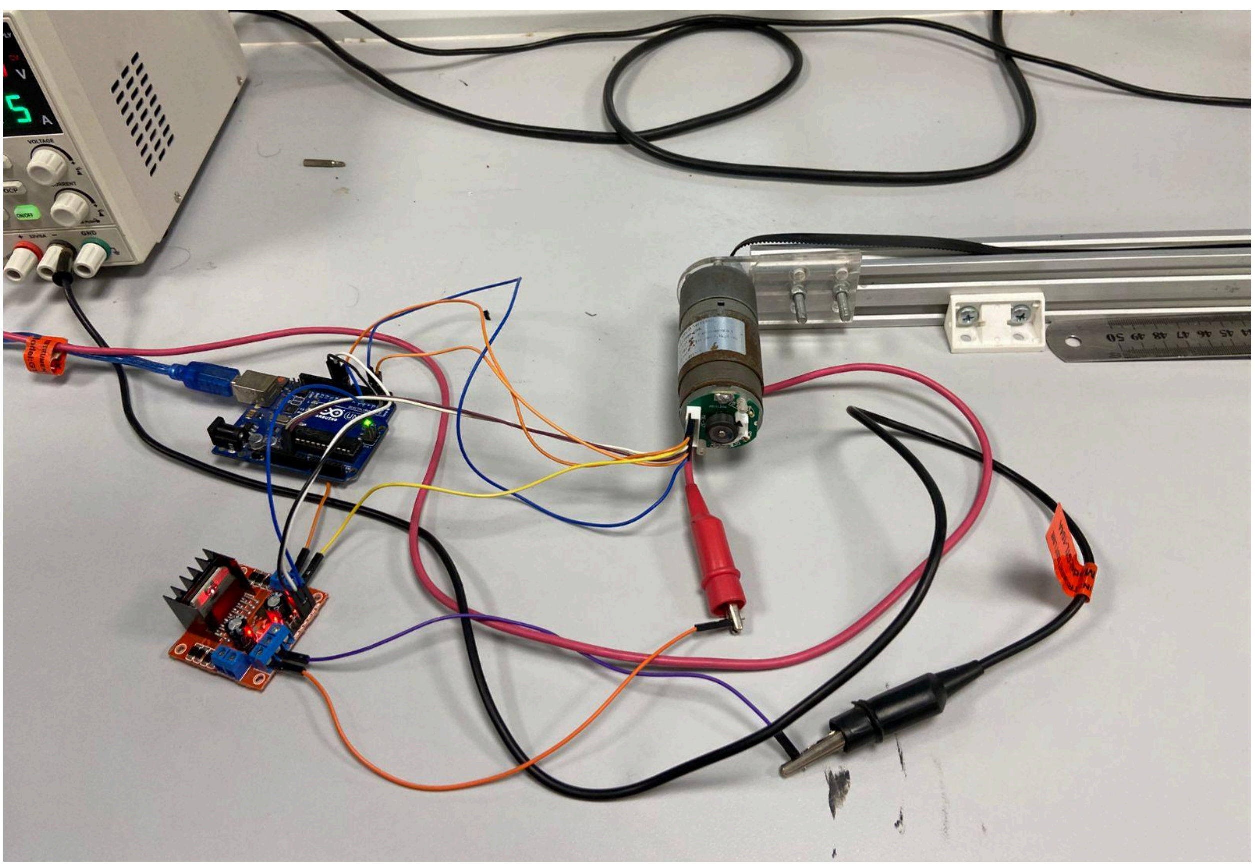


Mechanical Design:

1) SG555123000\10K DC motor with encoder.
2) 1 m 20*20 V-slot aluminum extrusion bar.
3) Mini-V gantry Plate.
4) Idler pulley plate.
5) Motor mount plate.
6) Mini V- wheel bearing kit.
7) Timing pulley GT2.
8) GT2 timing belt.
9) Smooth idler pulley kit.

Circuit Connection:





Encoder:

VCC-> +5v on motor driver L298

GND-> GNDonmotordriver L298

Channel A-> Arduino pin 2

Channel B-> Arduino pin 3

Motor:

M+->Connected to OUT 1 on motor driver L298

M--> Connected to OUT 2 on motor driver L298

Arduino:

Pin 2-> encoder A

pin 3-> encoder B

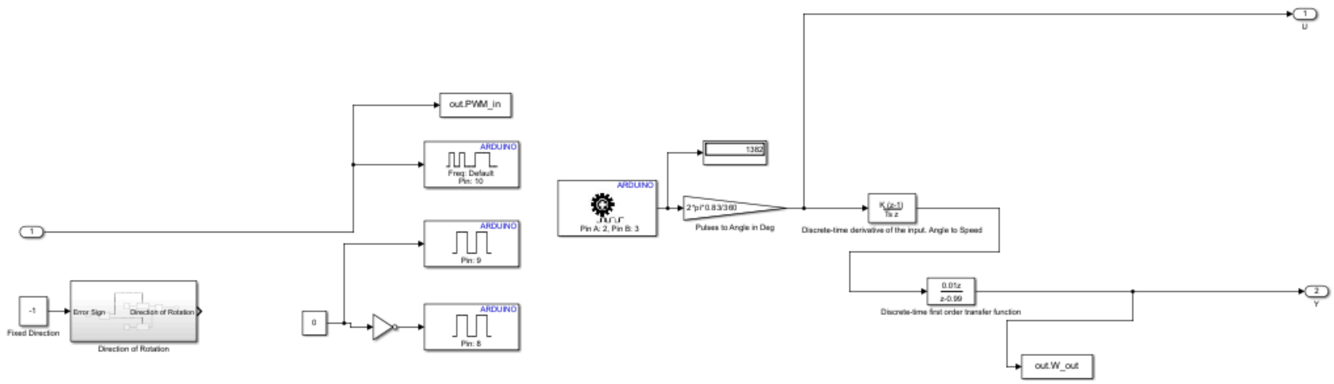
pin 8-> IN1 on motor driver

pin 9-> IN2 on motor driver

pin 10-> ENA on motor driver

GND->GNDonmotordriver

Simulink Model:

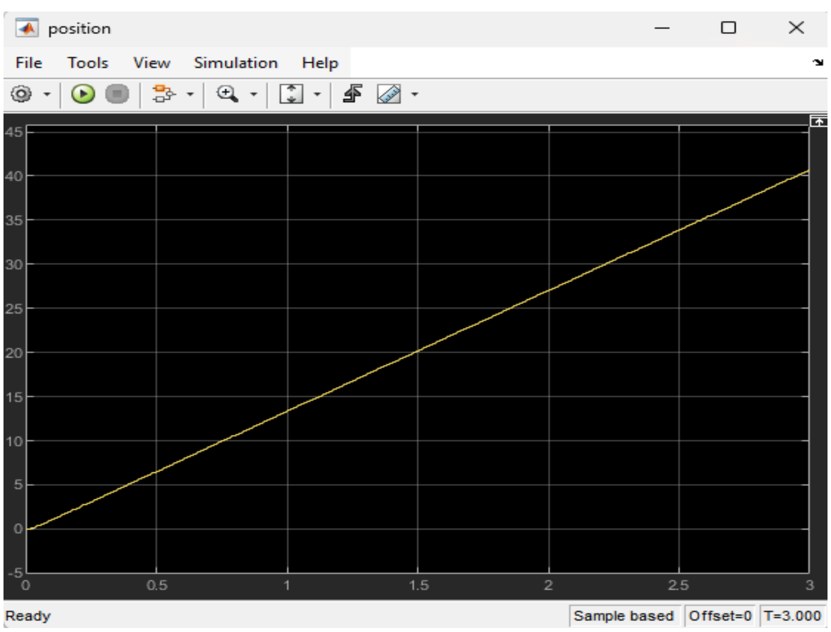


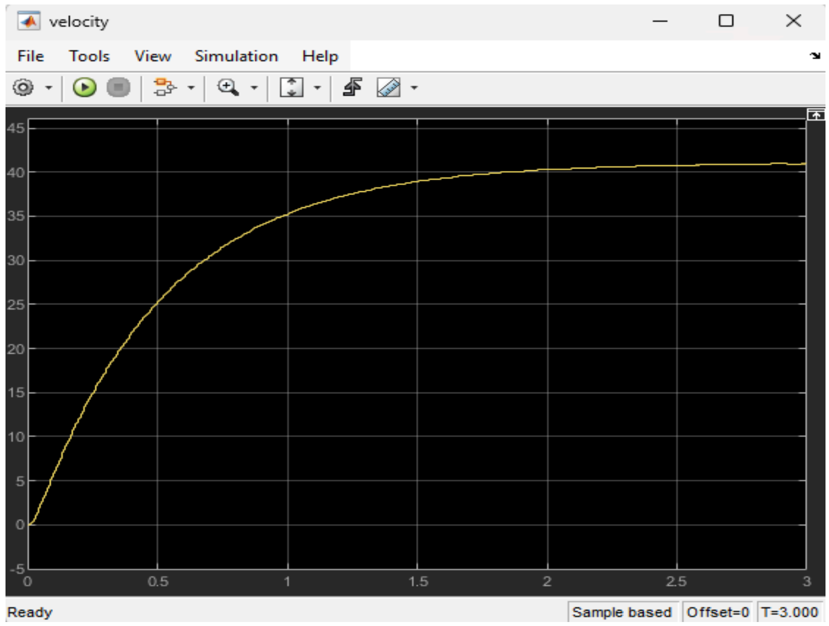
Data Collection:

We calculated the PPR by dividing the pulses over revolutions for sample time = 50 sec, we made several trials and we got PPR = 360

We got Ra by resisting the motion of the motor and stopping it where I=2.6 A and V=12 Volts thus Ra= V/I = 12/2.6=4.62 Ohm

Before parameter estimation and tuning:





The data is documented in the attached excel sheet.

As shown in the velocity curve:

the settling time = 2 sec

thus tau = 2/4=0.5

thus Ki/Kp=0.5

Parameter Estimation:

We estimated the equivalent viscous friction, armature resistance, back emf constant, armature inductance, and equivalent moment of inertia.

J = equivalent moment of inertia

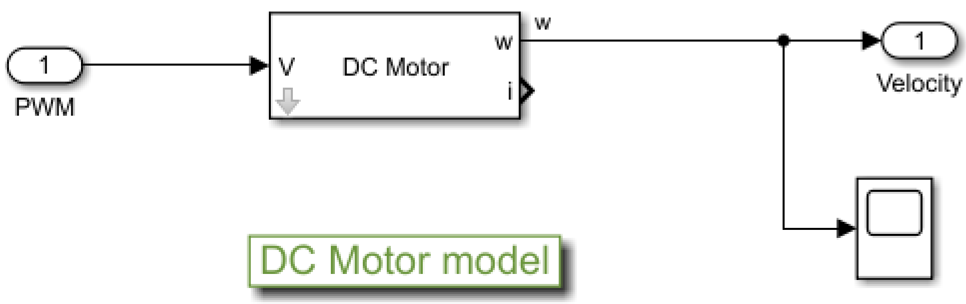
K_bh =back emf constant

b = equivalent viscous friction

La = armature inductance

Ra = armature resistance

DC motor model:

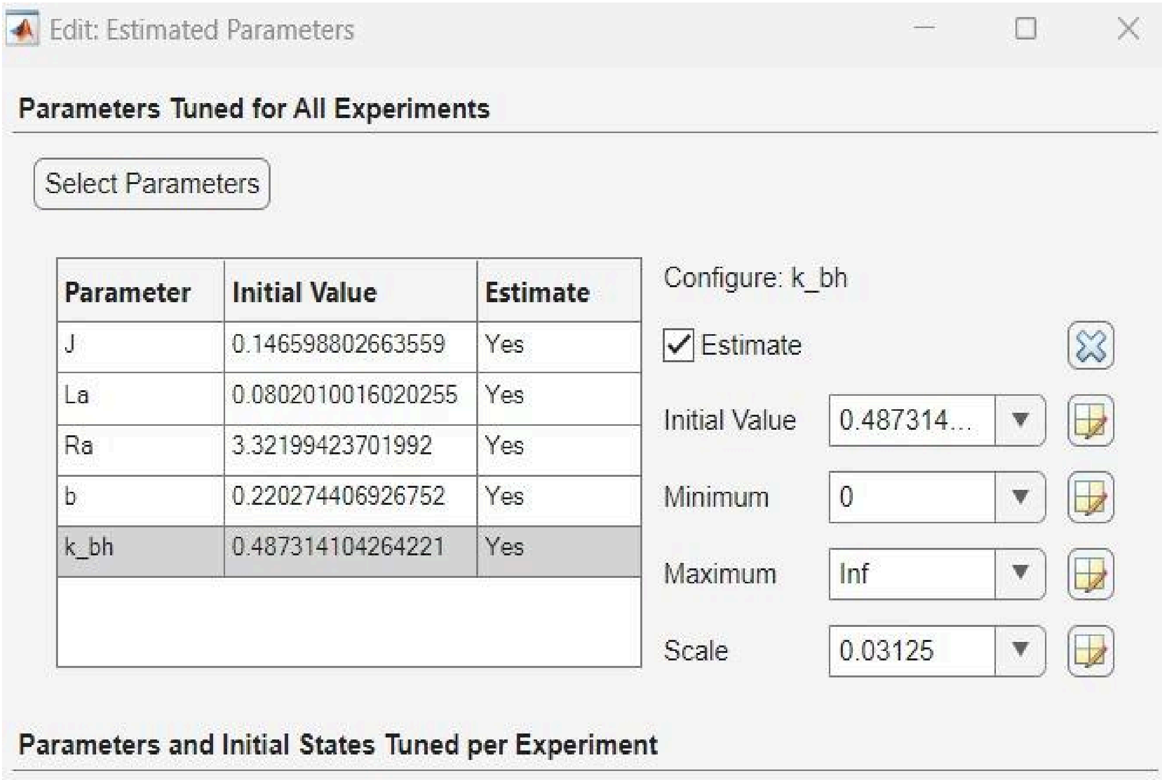


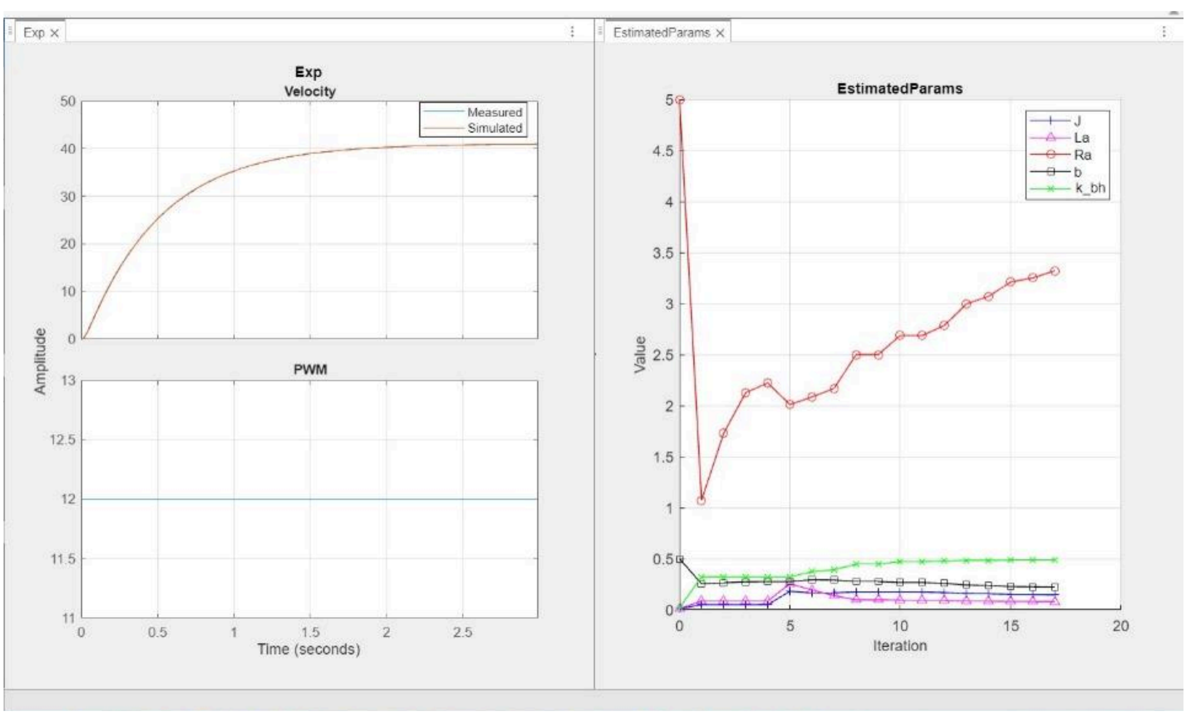

```
initialization.m
Ra = 5;
k_bh = 0.3;
b = 0.01;
La = 0.1 ;
J = 0.1;
```

We compared the experimental data for velocity and PWM for 3 sec time showed in the .mat file and excel sheet file with the simulation data and made parameter estimation as follows:

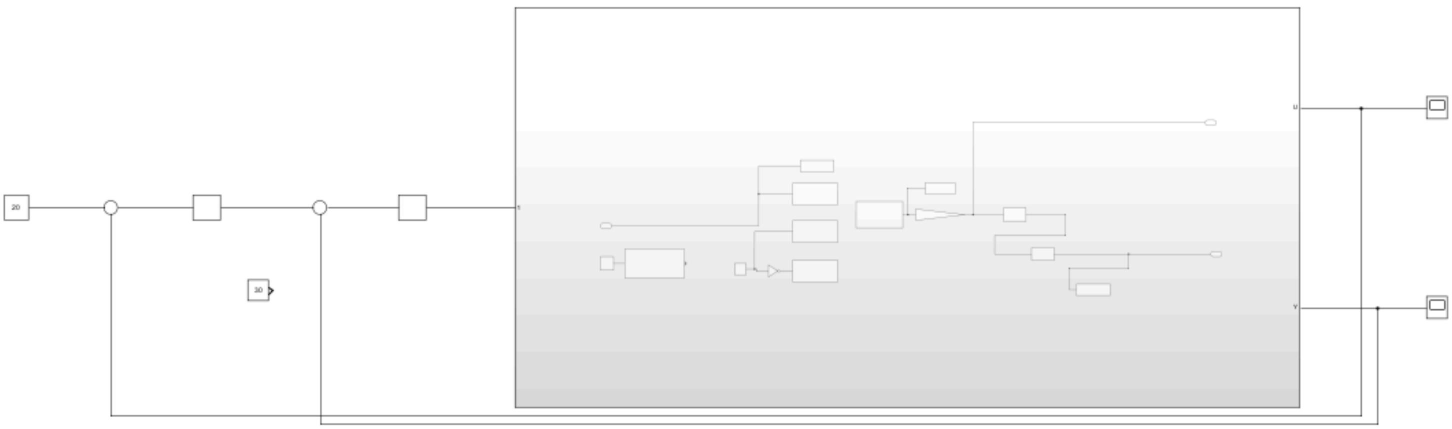
Estimation Progress Report

Iteration	F-count	Exp (Minimize)
0	11	432.5197
1	22	370.2485
2	33	44.2998
3	44	11.5087
4	55	10.5603
5	66	3.6624
6	77	0.3973
7	88	0.1025

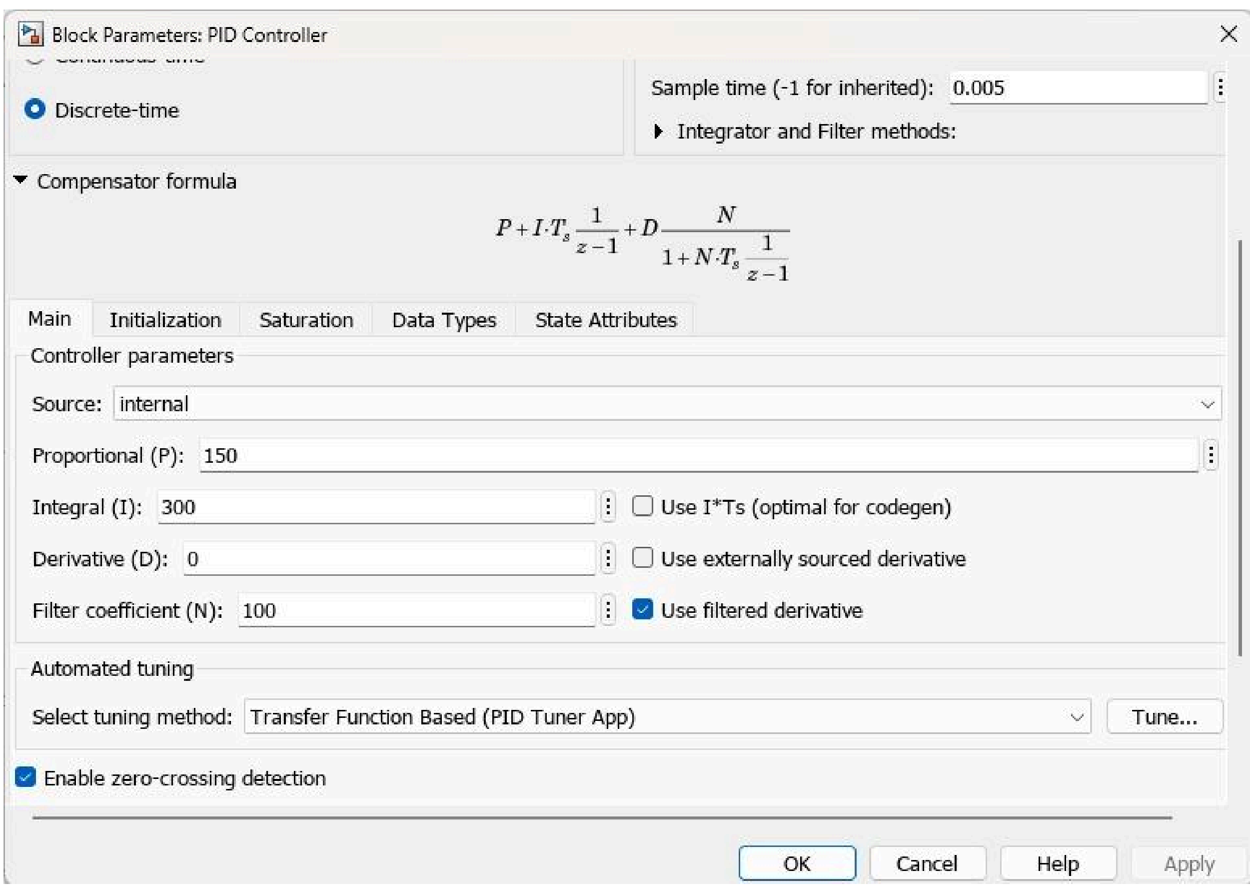

Edit: Estimated Parameters
Parameters Tuned for All Experiments
Select Parameters
Parameter | Initial Value | Estimate
J | 0.146598802663559 | Yes
La | 0.0802010016020255 | Yes
Ra | 3.32199423701992 | Yes
b | 0.220274406926752 | Yes
k_bh | 0.487314104264221 | Yes
Configure: k_bh
Estimate
Initial Value 0.487314...
Minimum 0
Maximum Inf
Scale 0.03125
Parameters and Initial States Tuned per Experiment

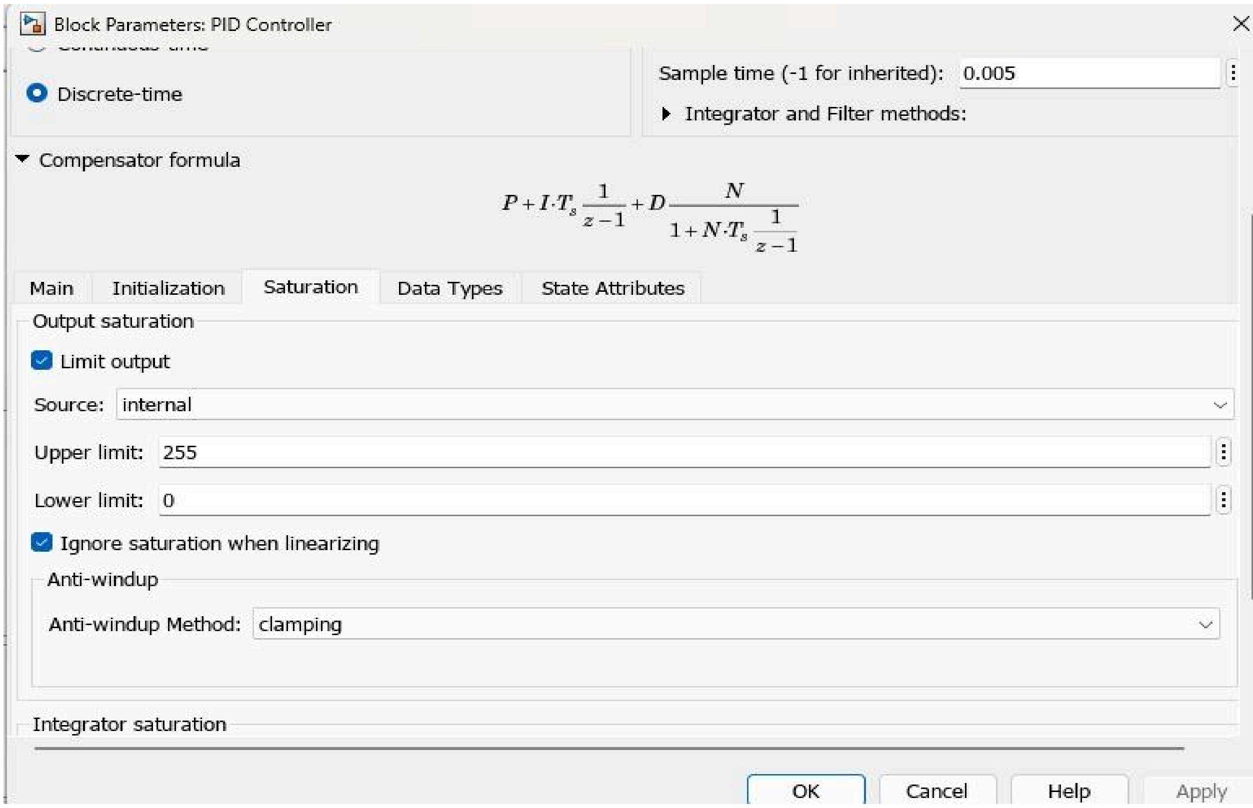

Exp
EstimatedParams
Exp
Velocity
Measured
Simulated
PWM
Amplitude
Time (seconds)
EstimatedParams
J
La
Ra
b
k_bh
Value
Iteration

Cascaded PID Tuning

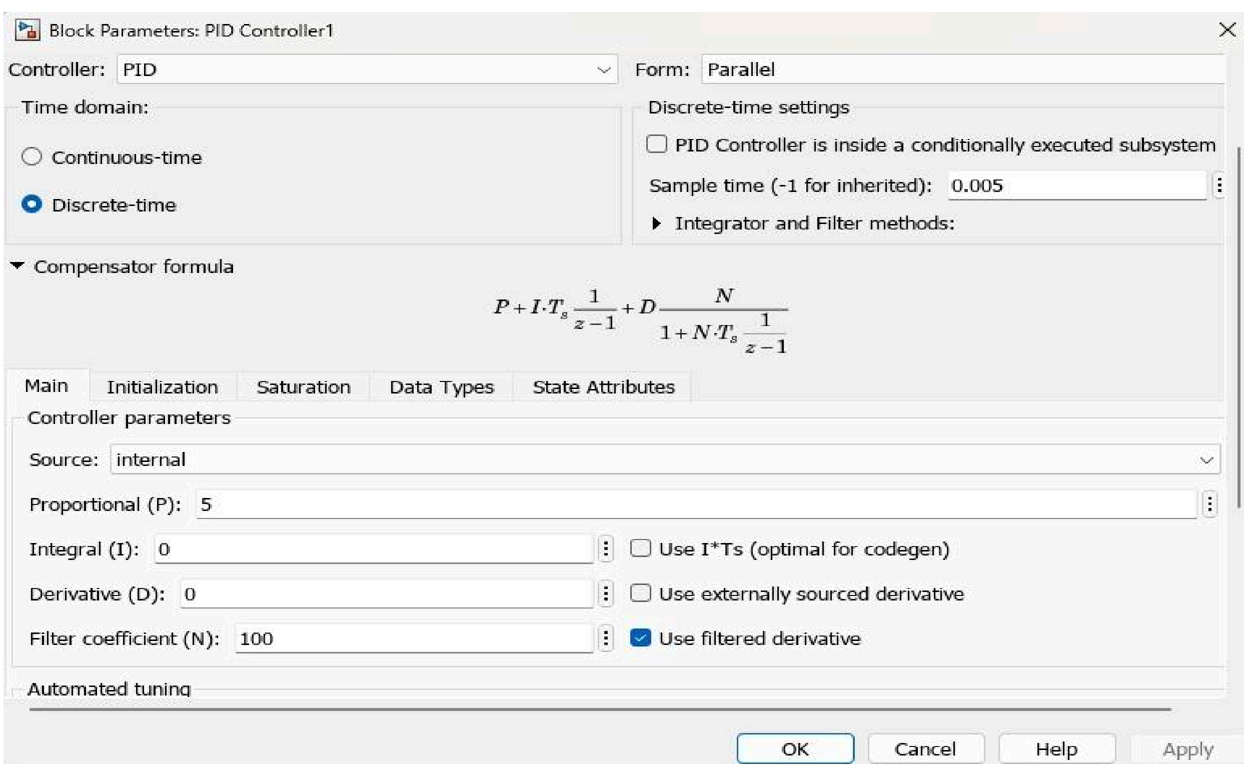


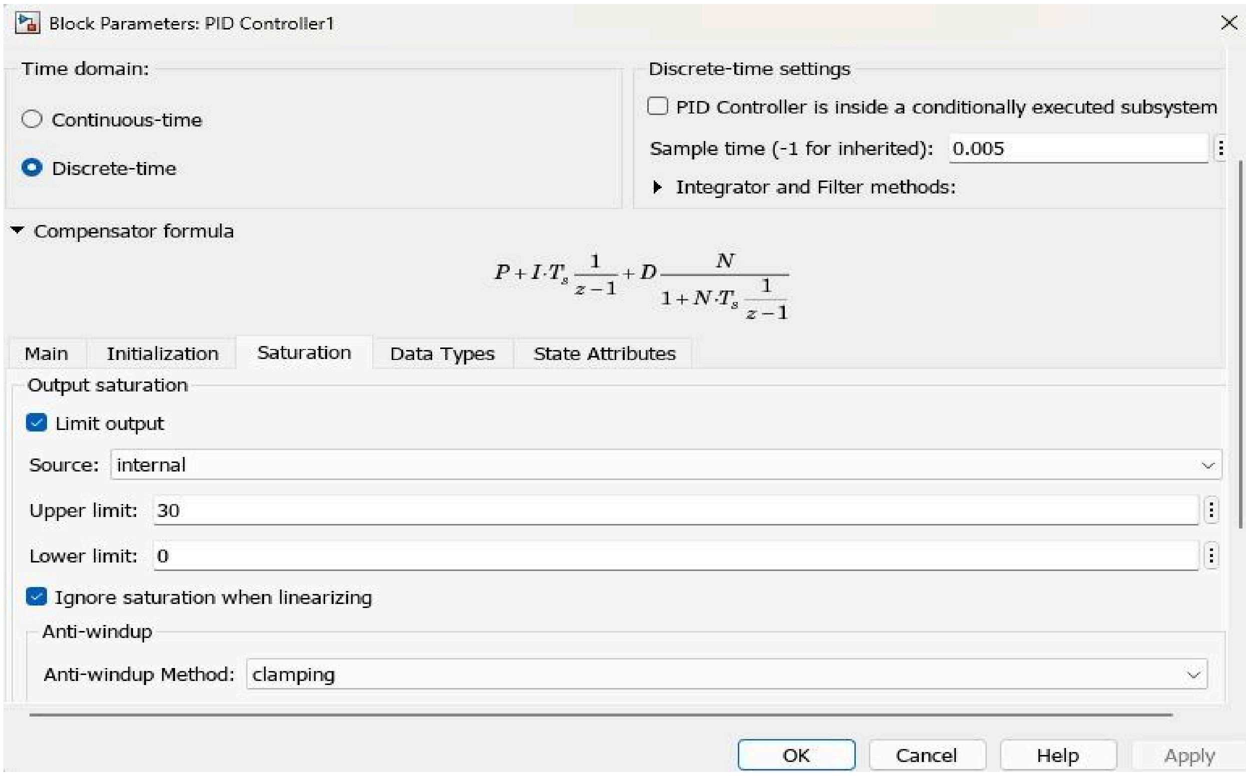
Velocity loop:



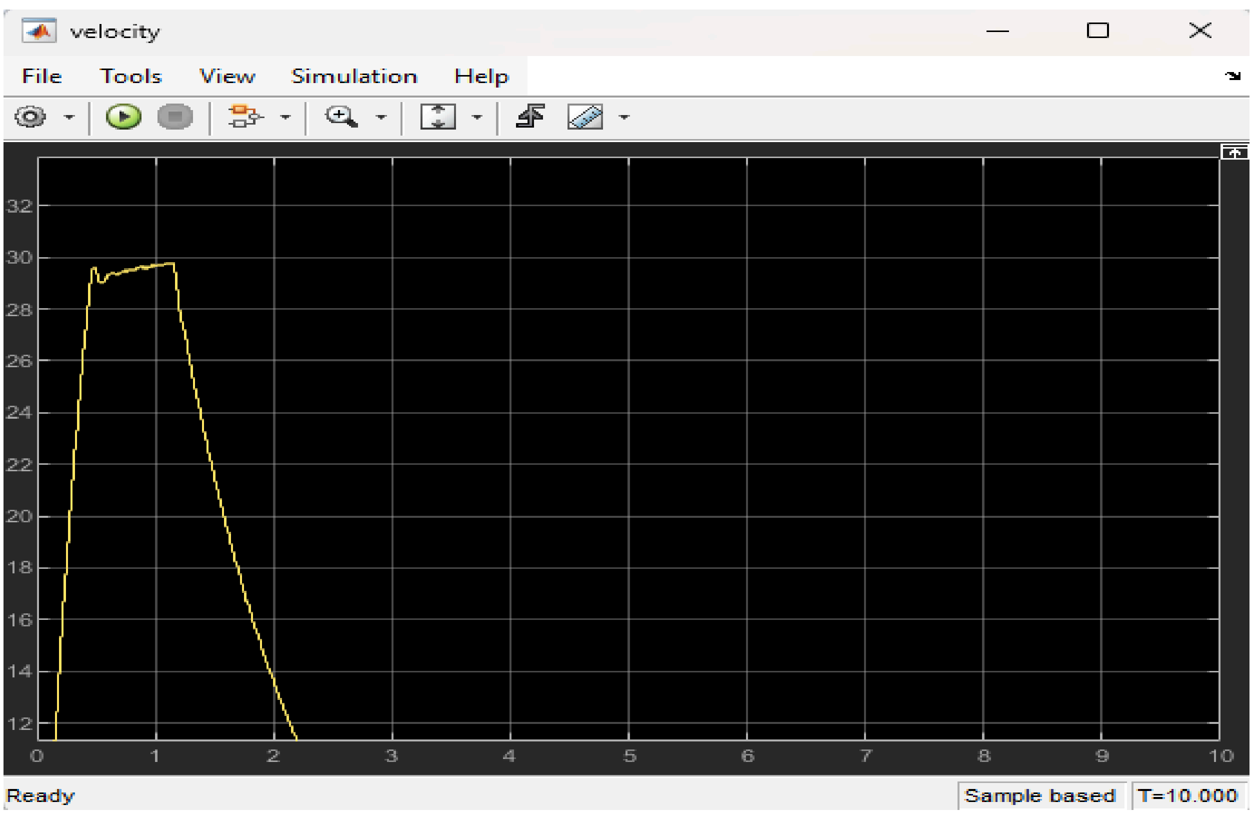


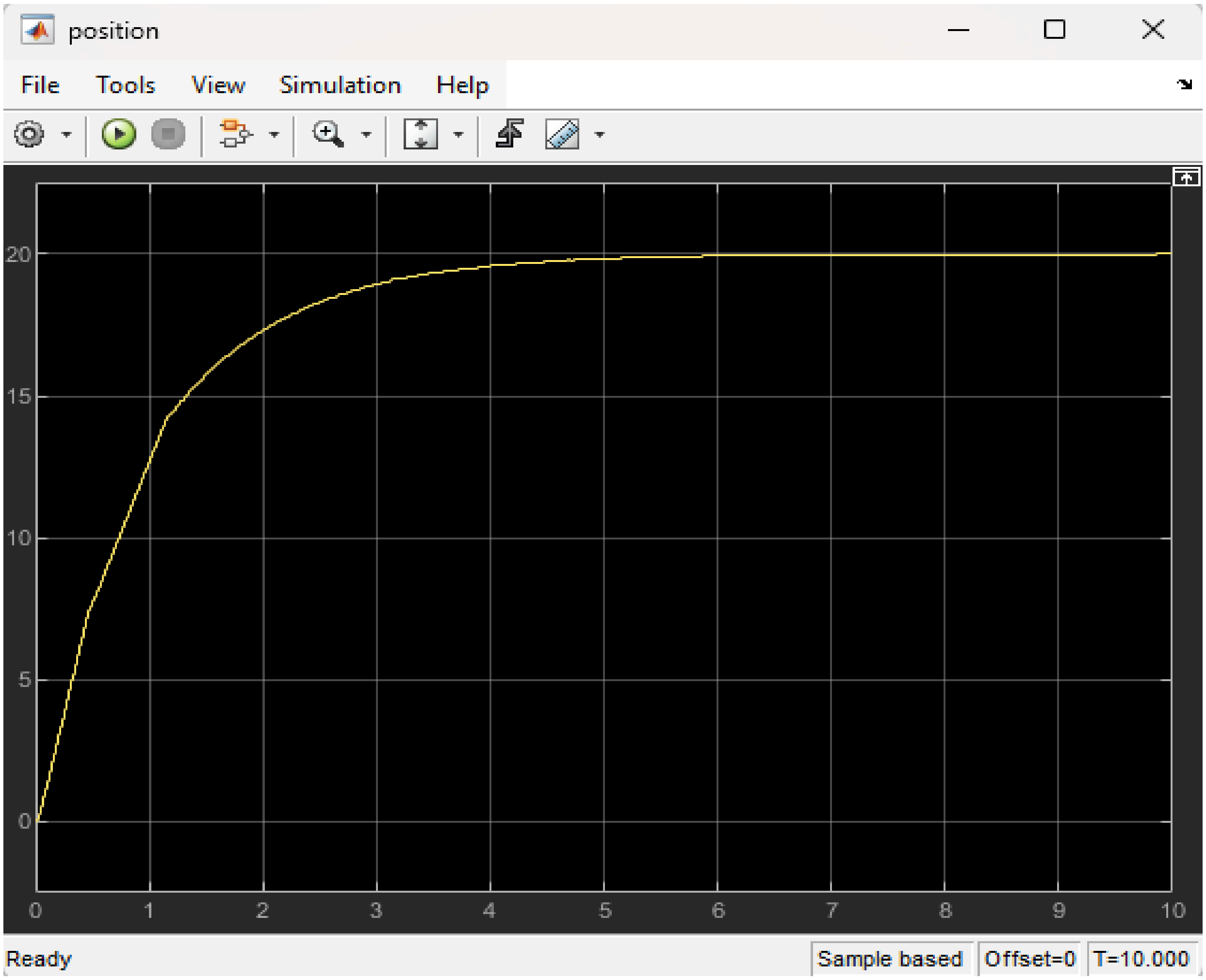
Position loop:





Model Response after tuning:





From the velocity response graph, the settling time is reduced from 2 sec to 0.42 sec and the position error becomes 2.5%, thus for 20 cm input, the cart moves 20.5 cm.

Discussion

This project had some issues summed up as follows: there were setup, control task, and parameter estimation issues.

Setup: there were wiring connections issues as many components were not working and it was hard to find the working jumpers, arduino, h-bridge and motor. Moreover, the motors were not working properly except for the one we used.

Tuning: direction block was not working so we reversed the direction of the cart.

Parameter estimation: ratio of Ki\Kp= 41 was not used for its purpose; instead, we used ratio 0.5 after several iterations for the best response.

Results

Controlling movement of the linear cart using cascaded PID the controller was able to effectively reduce the overshoot and settling time of the system, achieving a settling time of 0.42 seconds. These results indicate that the cascaded PID controller is an effective control strategy for movement control of linear carts, as it allows for precise and accurate positioning of the cart. The results of the experimental data showed that the linear cart was able to reach and maintain the desired position with a high degree of accuracy (error 2.5%).

Conclusion

We managed to successfully build a mechatronic linear servo system after correctly forming the setup, building the simulink model, initializing parameters, performing parameter estimation, making Cascaded PID tuning, and consequently the desired position was achieved accurately.

Work contribution:

Rofida Aboelwafa 201-901-861	Simulink Model + PID tuning + Report
Yasmina Elhefnawy 201-900-8...	Simulink Model + PID tuning + Report
Rehand Yassin 202-000-770	Circuit Wiring + Parameter Estimation
Ali Toema 201-900-863	Mechanical Setup + Parameter Estimation
Yara Abdullah 202-000-629	Data Collection

Drive link:

https://drive.google.com/drive/folders/10I72-czeZ0ZP1vjsDGWdKNKpKrX1tHpA?usp=sharing